

# Ramchander Rao Bhaskara

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## Academic Appointments

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| <b>Dartmouth College</b><br>Department of Computer Science<br><i>Postdoctoral Associate</i>   <i>Advised by Prof. Adithya Pediredla</i> | Dec 2025 – present |
| <b>Texas A&amp;M University</b><br>Department of Aerospace Engineering<br><i>Research Engineer</i>                                      | Sep – Dec 2025     |

## Education

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| <b>Texas A&amp;M University</b><br>PhD in Aerospace Engineering   <i>Advised by Prof. Manoranjan Majji</i>  | 2021 – 2025 |
| <b>Texas A&amp;M University</b><br>M.S. in Aerospace Engineering   <i>Advised by Prof. Manoranjan Majji</i> | 2019 – 2021 |
| <b>National Institute of Technology, Trichy</b><br>BTech in Instrumentation and Control Engineering         | 2013 – 2017 |

## Awards

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| <b>Future Leaders in Aerospace Symposium</b><br>Smead Aerospace Engineering Sciences, CU Boulder   <i>Invited participant</i>                                                                                                              | 2026        |
| <b>Best Talk Award</b> <a href="#"></a><br>Postdoctoral Research Day, Dartmouth College and University of New Hampshire                                 | 2026        |
| <b>Excellent Reviewer</b><br>Journal of Guidance, Control, and Dynamics (JGCD)                                                                                                                                                             | 2025        |
| <b>Guidance, Navigation, and Control (GNC) Graduate Award</b> <a href="#"></a><br>American Institute of Aeronautics and Astronautics   <i>3,500 USD</i> | 2024        |
| <b>Graduate Excellence Fellowship</b><br>Texas A&M University   <i>5 × 1,000 USD</i>                                                                                                                                                       | 2020 – 2024 |
| <b>Travel Award</b><br>Department of Aerospace Engineering, Texas A&M University   <i>2 × 500 USD</i>                                                                                                                                      | 2023 – 2024 |
| <b>Finalist, GNC Graduate Student Paper Award</b><br>AIAA SciTech Forum   <i>500 USD</i>                                                                                                                                                   | 2023        |
| <b>Best Student Research Paper Award</b> <a href="#"></a><br>Digital Avionics Systems Conference (DASC)   <i>500 USD</i>                                | 2022        |
| <b>Travel Grant</b><br>Office of Graduate and Professional Studies, Texas A&M University   <i>500 USD</i>                                                                                                                                  | 2022        |
| <b>ASIE Scholarship</b><br>American Society of Indian Engineers and Architects, Houston   <i>1,000 USD</i>                                                                                                                                 | 2022        |
| <b>NASA TechLeap Prize — GNC Lead</b> <a href="#"></a><br>NASA Flight Opportunities Program   <i>100,000 USD towards flight experiment</i>              | 2021        |
| <b>RECT Silver 72 Scholarship</b><br>National Institute of Technology, Trichy   <i>15,000 INR per year</i>                                                                                                                                 | 2015 – 2017 |

## IIT Madras Summer Research Fellowship

2016

Department of Aerospace Engineering, IIT Madras | *Granted but declined*

## Other Research Experience

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### Cornell University

2025

Mechanical and Aerospace Engineering | *SmallSat Mission Design School*

### NASA Jet Propulsion Laboratory

2022, 2023

Mobility and Robotic Systems Section | *Research Intern*

### iRunway India (now UnitedLex)

2017 – 2019

Intellectual Property | *Associate (5G/LTE network architecture)*

## Publications

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### SDR-ToF: Million Optical Depth Samples per Second Using Software-Defined Radio [🔗](#)

Ramchander Rao Bhaskara\*, Dhawal Sirikonda\*, Omkar Vengurlekar\*, Juhyeon Kim, Joseph Lazarro, Suren Jayasuriya, Adithya Pediredla

*ACM Transactions on Graphics (SIGGRAPH Asia), 2026*

### ShutterEvents: Shutter Modulation for Static-Dynamic Scene Recovery [🔗](#)

Nevindu Batagoda\*, Ramchander Rao Bhaskara\*, Christian Metzler, Adithya Pediredla

*IEEE International Conference on Computational Photography (ICCP) 2026*

### Development and Validation of Velocimeter Lidar Simulator [🔗](#)

Ramchander Rao Bhaskara, Roshan T. Eapen, and Manoranjan Majji

*AIAA SciTech 2025*

### Quantized State Estimation for Linear Dynamical Systems [🔗](#)

Ramchander Rao Bhaskara, Manoranjan Majji, and Felipe Guzmán

*Sensors 2024*

### Icy Moon Surface Simulation and Stereo Depth Estimation for Sampling Autonomy [🔗](#)

Ramchander Rao Bhaskara, G Georgakis, J Nash, J Bowkett, M Cameron, A Ansar, P Backes, and M Majji

*IEEE Aerospace Conference 2024*

### Differentiable Rendering for Pose Estimation in Proximity Operations [🔗](#)

Ramchander Rao Bhaskara, Roshan T. Eapen, and Manoranjan Majji

*AIAA SciTech Forum 2023* | [Finalist, GNC graduate student papers](#)

### An FPGA Framework for Interferometric Vision-Based Navigation (iVisNav) [🔗](#)

Ramchander Rao Bhaskara, Kookjin Sung, and Manoranjan Majji

*41<sup>st</sup> Digital Avionics and Systems Conference 2022* | [Best student research paper](#)

### FPGA Hardware Acceleration for Feature-Based Relative Navigation Applications [🔗](#)

Ramchander Rao Bhaskara and Manoranjan Majji

*AAS/AIAA Astrodynamics Specialist Conference 2022*

### Vision and Inertial Sensor Fusion for Terrain Relative Navigation [🔗](#)

A. Verras, R. Eapen, A Simon, M Majji, Ramchander Rao Bhaskara, C. Restrepo, and R. Lovelace

*AIAA SciTech Forum 2021*

### Interferometric Vision-Based Navigation Sensor for Autonomous Proximity Operation [🔗](#)

Kookjin Sung, Ramchander Rao Bhaskara, and Manoranjan Majji

*39<sup>th</sup> Digital Avionics and Systems Conference 2020*

### Under Review

On the Low-Pass Filter Design for Digital Phasemeter

Ramchander Rao Bhaskara and Manoranjan Majji

Submitted to IEEE Transactions on Instrumentation and Measurement 2026

## Patents

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### MHz-Rate Optical Depth Imaging Using Software-Defined Radios

Adithya Pediredla, Ramchander Rao Bhaskara, Dhawal Sirikonda, Omkar Vengurlekar, Juhyeon Kim, Joseph Lazzaro, Suren Jayasuriya

U.S. Patent Application No.: 64/113,861

## Other Publications

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### Real-Time Signal Processing and State Estimation for Spaceflight Applications [↗](#)

Ramchander Rao Bhaskara

PhD Dissertation, supervised by Prof. Manoranjan Majji

### Hardware Implementation of Navigation Filters for Automation of Dynamical Systems [↗](#)

Ramchander Rao Bhaskara

Master's Thesis, supervised by Prof. Manoranjan Majji

### Searching for Ariel's Aliens with the CRISPI Mass Spectrometer SmallSat Flyby Mission [↗](#)

Zach Ulibarri, S Jamal, A Alex, L Arzoumanian, Ramchander Rao Bhaskara, A Chadwick, S Gayen et al.

American Geophysical Union 2025

### Precision Depth Sensing: Space Science to Robotics

Ramchander Rao Bhaskara

Guarini Poster Session, Dartmouth College, 2026

### Toward Development and Validation of Velocimeter LiDAR Simulator

Ramchander Rao Bhaskara and Manoranjan Majji

2024 AAS Guidance, Navigation and Control Conference., Breckenridge, Colorado

### NaRPA: Navigation and Rendering Pipeline for Astronautics [↗](#)

Ramchander Rao Bhaskara\*, Roshan T. Eapen\*, and Manoranjan Majji

Poster, Lunar Surface Innovation Consortium 2021

### Physics-Based Modeling of Selective Catalytic Reduction System and Reduced Order Dynamics [↗](#)

Ramchander Rao Bhaskara

Undergraduate Thesis, supervised by Prof. Umapathy Mangalanathan, NIT Trichy

## Talks

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### Precision depth sensing: space science and robotics

Future Leaders in Aerospace Symposium | *Invited talk*

University of Colorado Boulder | 2026

Postdoctoral Research Day

Dartmouth College | 2026

### On-Board Sensing and State Estimation for Opto-Mechanical Inertial Sensors

Starlink (SpaceX) | *hosted by Javier Vicens (GNC)*

Seattle | 2025

### Computer Vision for Icy-World Exploration

Machine-Ground Interaction Consortium (MaGIC) | *hosted by Prof. Dan Negrut*

UW Madison | 2025

### On Sensor Simulations for Planetary Exploration

Visual Computing Seminar | *Invited Talk*

Dartmouth College | 2025

### Sampling Autonomy for Europa Lander

Jet Propulsion Laboratory | *Intern Talk*

Pasadena, CA | 2023

### Study of Topology of Icy Moons

Jet Propulsion Laboratory | *Intern Talk*

Pasadena, CA | 2023

### Texas A&M ScORE: Space Object Rendering Engine

Lunar Surface Innovation Consortium | *Symposium*

Johns Hopkins University | 2021

## Academic Service

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### International Conference on Computational Photography (ICCP)

2027

Website chair

### American Institute of Aeronautics and Astronautics (AIAA)

2024 – Present

Sensor Systems and Information Fusion (SSIF) | *Technical Committee*

### AIAA SciTech Forum

2025

Guidance, Navigation, and Control (GNC-44) | *Session Chair*

### AIAA SciTech Forum

2025

Sensor and Algorithm Analysis (SEN-02) | *Session Chair*

Hosted colloquium talks: Georgios Georgakis (JPL, 2026), Jimin Wu (Rice University, 2026)

Hosted lab visits/demos: Whitney Mason (DARPA), Xia Zhou (Columbia University), Daphna Weinshall (Hebrew University); Anthony Rizzo, Bryce Ferguson, Alberto Quattrini Li, Jonathan Chipman (Dartmouth); Andrew Morgan (RAI Institute), Dartmouth Class of 1962 (alumni).

Demos at ICCP 2026, Texas Regional Robotics Symposium - 2023, 24.

## Reviews

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Journal of Astronautical Sciences (JAS) (3)

2024 – 26

IEEE Transactions on Computational Imaging (1)

2026

Journal of Guidance, Control, and Dynamics (JGCD) (1)

2025

IEEE Control Systems Letters (1)

2025

American Control Conference (4)

2022 – 25

AIAA SciTech Forum (8)

2025 – 26

Transactions on Computers (1)

2023

## Teaching

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### AERO 423: Orbital Mechanics

2024, 2025

Texas A&M University | *Teaching Assistant* [🔗](#)

*Student feedback: "Your support and guidance during the courses have been incredibly helpful, and I truly appreciate you cheering me on as I prepare for my defense. It means a lot to have mentors like you."*

### Digital Signal Processing

2023

Texas A&M University | *Department Tutorial Lecture*

## Mentoring

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### Ankit Burudgunte

2026

Rice University | *Undergraduate research (NSF REU), optical backscatter communication*

### Dhawal Sirikonda

2026

Dartmouth College | *PhD project, digital signal processing*

### Nevindu Batagoda

2026

Dartmouth College | *PhD project, event sensors*

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| <b>Peiran Li</b><br>Dartmouth College   <i>MS research, optics</i>                                                                           | 2026 |
| <b>Omkar Vengurlekar</b><br>Arizona State University   <i>PhD project, audio signal processing</i>                                           | 2026 |
| <b>Team Lunatyx</b><br>PI for Lunar Autonomy Challenge 2025   <i>2 graduate and 3 undergraduate students, visual navigation on Moon</i>      | 2025 |
| <b>Omar Mohmand &amp; Marco Peredo</b><br>Texas A&M University   <i>Undergraduate research: Trajectory design for Mars/Europa rendezvous</i> | 2024 |

## Volunteering

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| <b>Dartmouth College</b><br>Annual Science Day   <i>Demonstrations to K-12 students</i>                                                                 | 2026        |
| <b>Montshire Museum of Science - Vermont</b><br>Astronomy Day   <i>Speaker, Astronomer in the House</i>                                                 | 2026        |
| <b>Texas A&amp;M University</b><br>Aerospace Engineering Graduate Student Association (AEGSA)   <i>Professional development chair</i> <a href="#">↗</a> | 2023 – 2025 |
| <b>Texas A&amp;M University</b><br>Science and Astronomy Festival   <i>Volunteer, outreach demonstrations</i>                                           | 2022 – 2025 |
| <b>Bhumi NGO</b><br>Renuka High School, Bangalore   <i>Volunteering physics teacher</i>                                                                 | 2017 – 2019 |

## Software

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| <b>GUISS:</b> Graphical Utility for Icy Moon Surface Simulations <a href="#">↗</a> | 2024 |
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